

Sample End Semester Question Paper - 2025

Subject: Probability and Statistics | Code: DS301 | Semester: 3

Branch: Data Science | Regulation: R20 (sample) | Time: 3 Hours | Max

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. State one property of the normal distribution.
2. Define a hypothesis.
3. What is the null hypothesis?
4. State the mean of the binomial distribution.
5. Define the p-value.
6. Define the arithmetic mean.
7. What is a probability distribution?
8. State Bayes' theorem.
9. Define a random variable.
10. What is a sampling frame?

PART B - Long questions (6 x 13 = 78 Marks)

1. Explain measures of central tendency: mean, median and mode with examples.
2. Explain binomial and Poisson distributions with applications.
3. State and apply Bayes' theorem with an example.
4. Explain sampling techniques and sampling error.
5. Explain estimation: point and interval estimation with examples.
6. Explain hypothesis testing: z-test for a population mean with an example.